

*"99.9% of science is boring."
- Makise Kurisu*

PRECALCULUS

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2 GRAPHICAL TRANSFORMATIONS

2.1 Introduction To Graphical Transformations

Definition 2.1 (Graphical Transformations). A mathematical operation that alters the position, size or orientation of a function's graph on a coordinate plane, without changing its basic shape.

2.1.1 Vertical

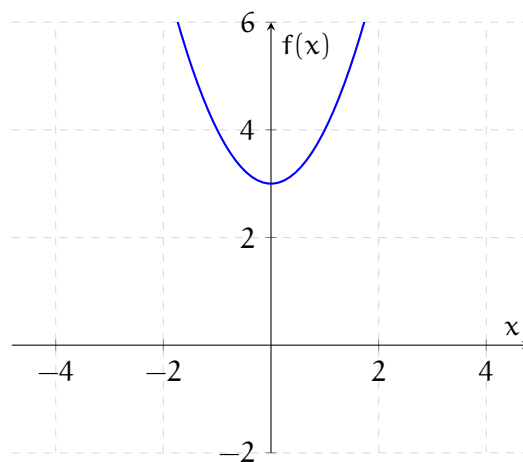
Definition 2.2 (Vertical Transformation).

$$f(x) + k \quad \text{Shift Up}$$

$$f(x) - k \quad \text{Shift Down}$$

Example 2.1 (Vertical Transformation).

$$f(x) = x^2 + 3$$



Note: Key points are $(1, 1)$, $(0, 0)$, $(-1, 1)$.

2.1.2 Horizontal

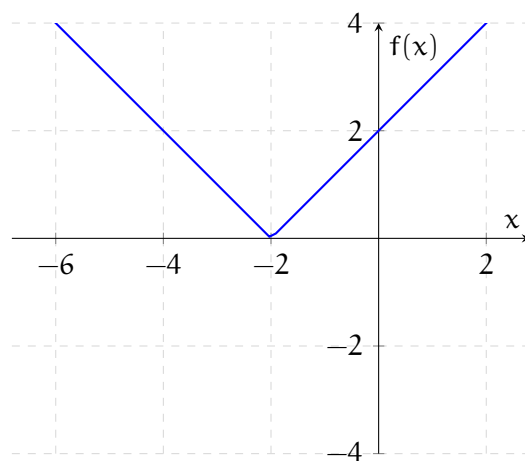
Definition 2.3 (Horizontal Transformation).

$$f(x + h) \quad \text{Shift Left}$$

$$f(x - h) \quad \text{Shift Right}$$

Example 2.2 (Horizontal Transformation).

$$f(x) = |x + 2|$$



Note: Key points are $(1, 1)$, $(0, 0)$, $(-1, 1)$.